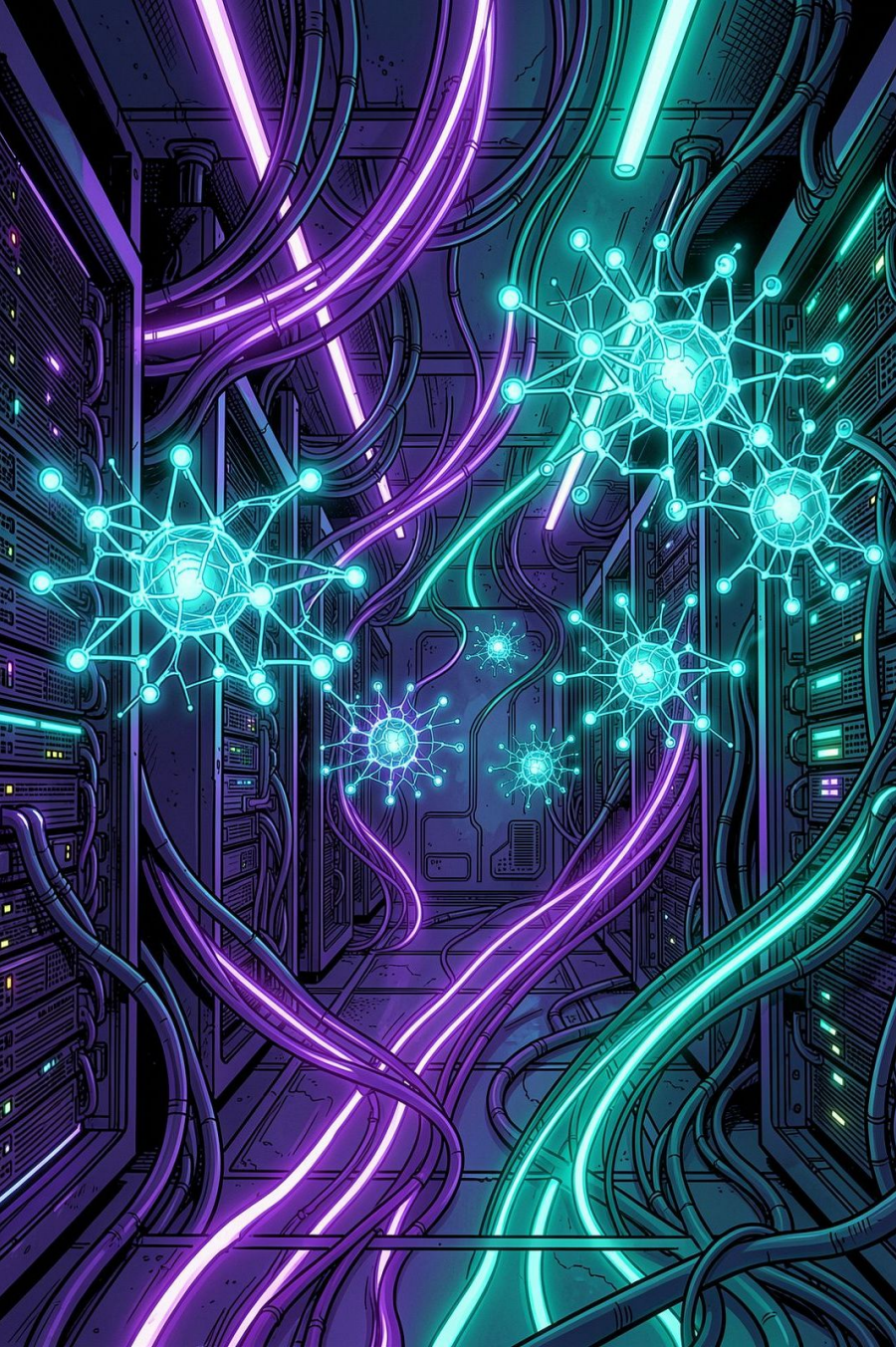




Corporate Onboarding & Training Intelligence Hub

Multi-Modal Agentic RAG with Division RBAC & RAGAS Observability

ENTERPRISE AI PLATFORM

V1.0 — PITCH DECK

The Knowledge Crisis in Enterprise

Knowledge Silos

Training content trapped in PDFs and video recordings — unsearchable, unindexed, and inaccessible at scale.

Security Leaks

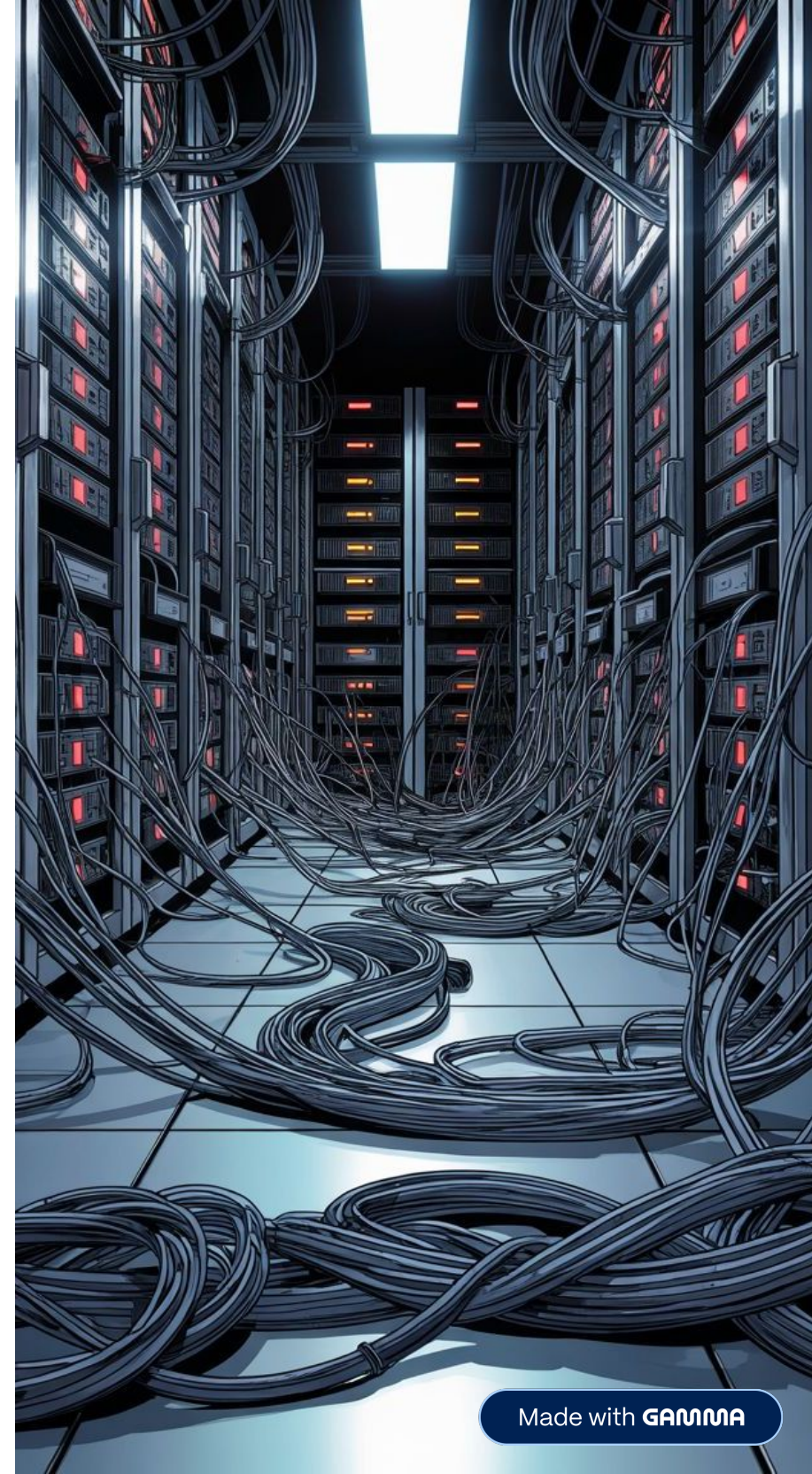
Cross-division data exposure from flat RAG architectures with no query-level access control.

Uncontrolled LLM Costs

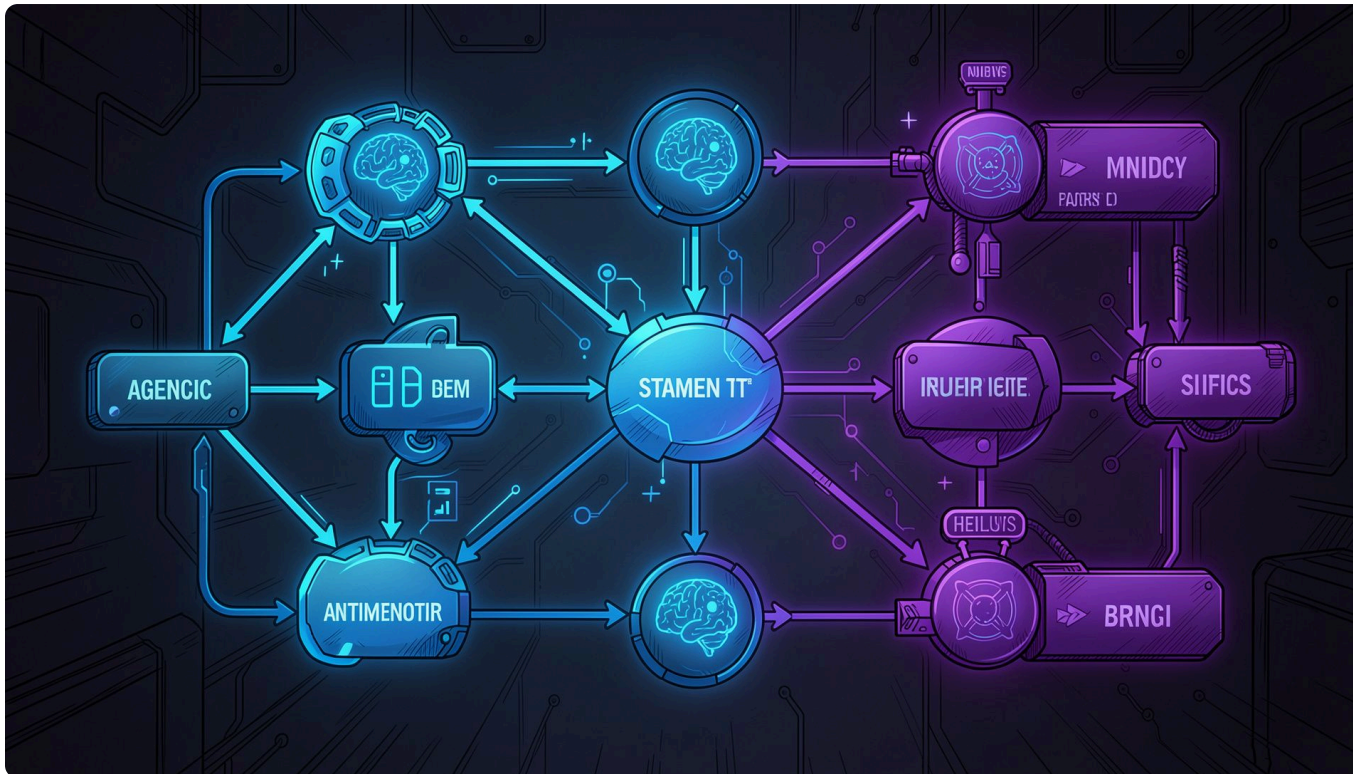
Bloated context windows and redundant inference driving unsustainable token expenditure.

Untrusted Search

Hallucinated answers and low-confidence retrieval eroding user trust in AI-assisted onboarding.



The Platform Solution



LangGraph Agentic RAG State Machine

Stateful orchestration of retrieval, routing, and generation — with explicit decision gates at every step.

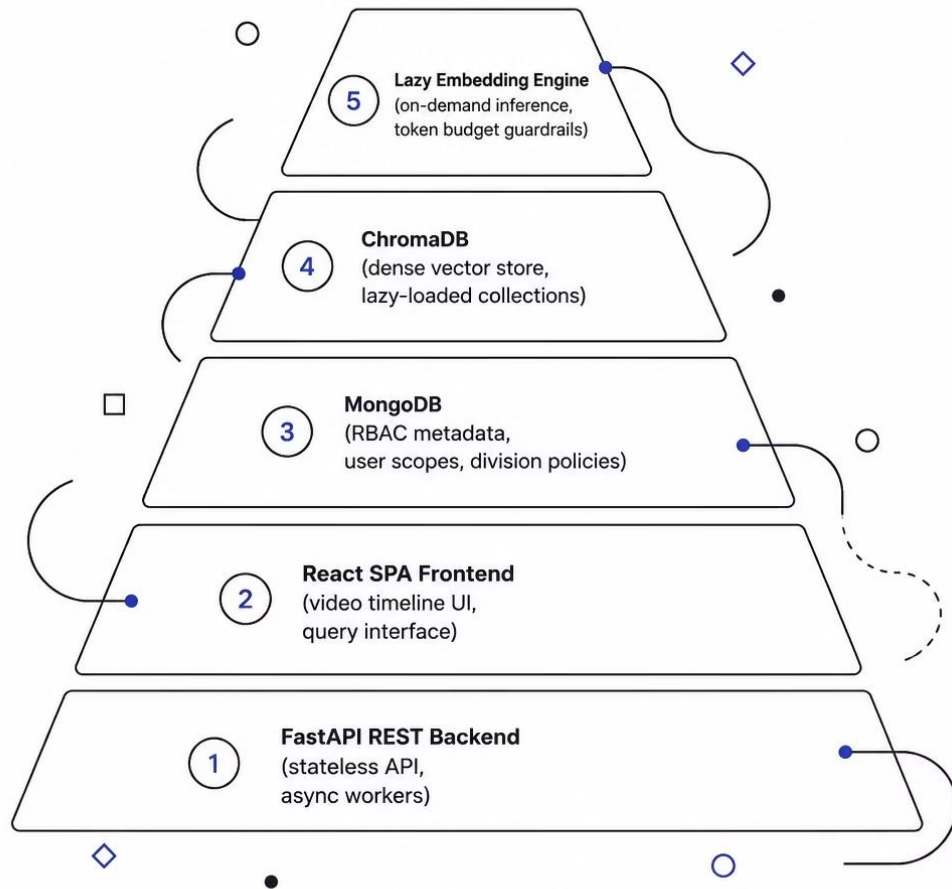
Query-Level Division RBAC

Every query is scoped to `division` and `businessLine` before vector search executes. No leakage, by design.

Whisper Media Ingestion + Semantic Caching

Audio/video transcribed via Whisper, embedded lazily, and cached at `user_email:query` granularity — cutting redundant LLM calls by 90%.

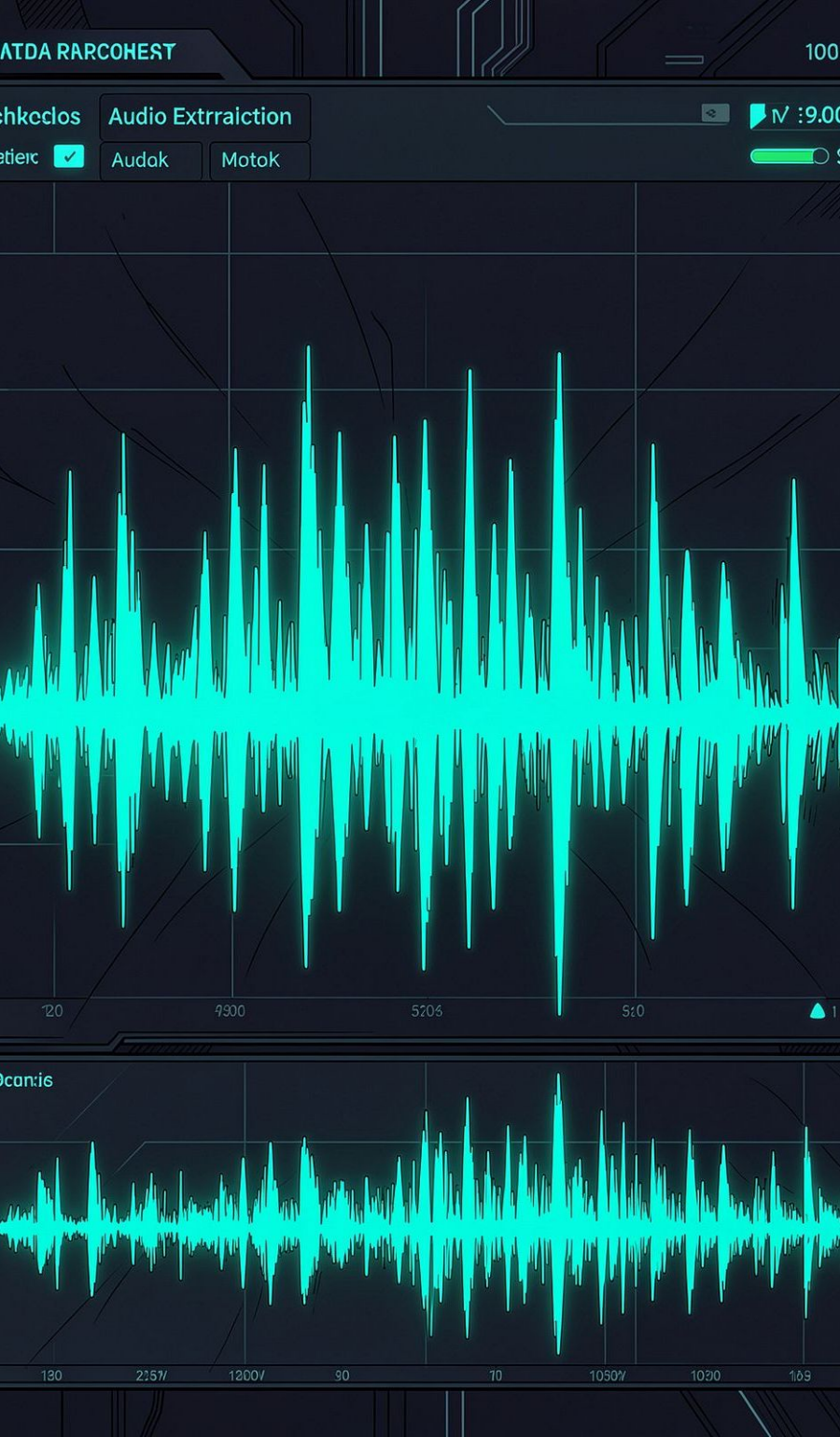
System Architecture



Layered for Scale & Security

The architecture separates concerns across five discrete layers — each independently scalable and auditable.

- **FastAPI** handles async request routing with sub-10ms overhead
- **MongoDB** stores RBAC policies, user scopes, and division metadata
- **ChromaDB** manages lazy-loaded vector collections per business unit
- **Lazy Embedding Engine** defers inference until query-time, slashing idle compute



Multi-Format Ingestion Pipeline

1

MoviePy Extraction

Audio pulled from video in **sub-0.5 seconds** — no transcoding overhead.

2

Whisper Transcription

OpenAI Whisper converts speech to text with speaker diarization support.

3

Paragraph Aggregation

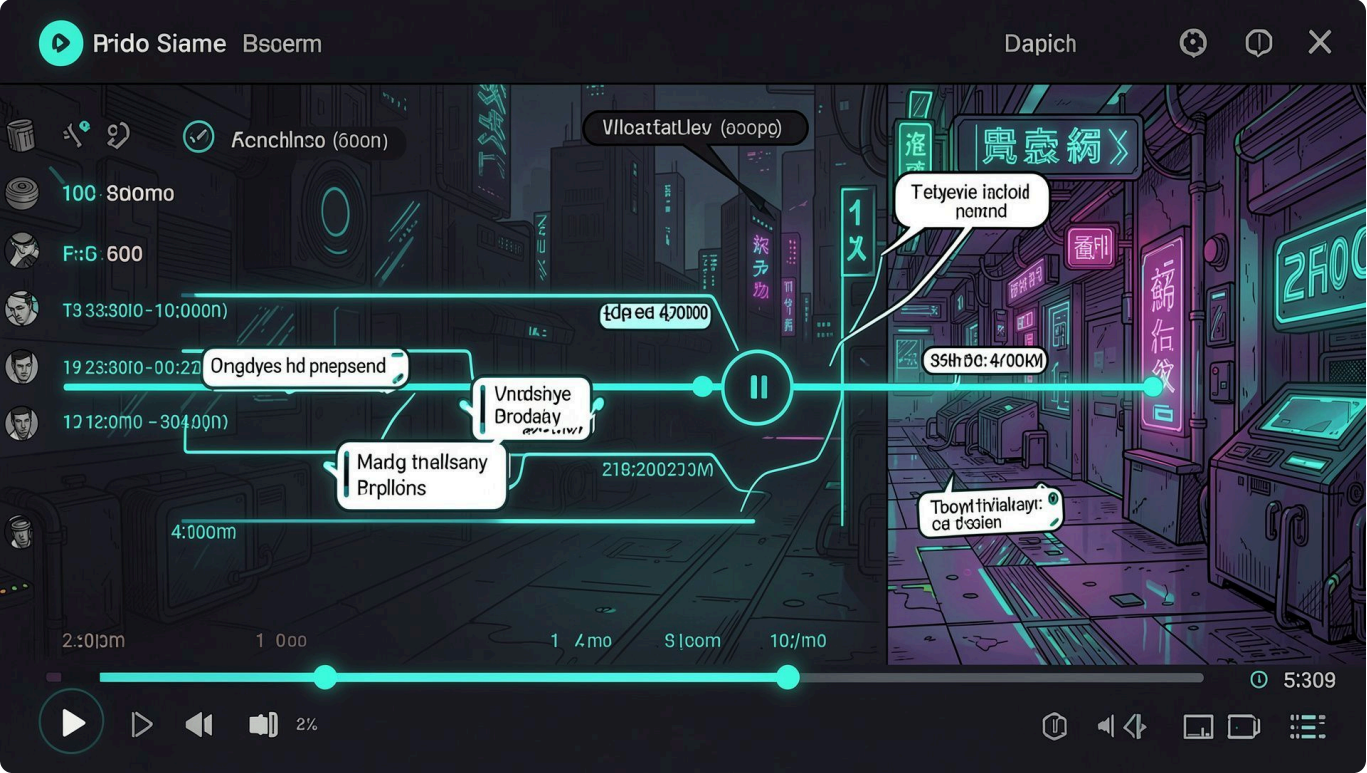
30–60 second timestamp windows merged into semantic chunks — **60%+ DB savings**.

4

Lazy Embedding

Vectors generated on-demand, not at ingest — eliminating cold-storage waste.

Interactive Video Timeline UI



Precision-Linked Retrieval

Users don't just get an answer — they get the *exact moment* in the training video where that answer was spoken.

Track Badges

Video and audio source indicators displayed inline with retrieved segments.

Merged Intervals

Continuous spans (e.g., 00:00–04:10) collapsed into single clickable regions.

Hover Speech Tooltips

Transcript snippets surface on hover — no playback required for quick scanning.

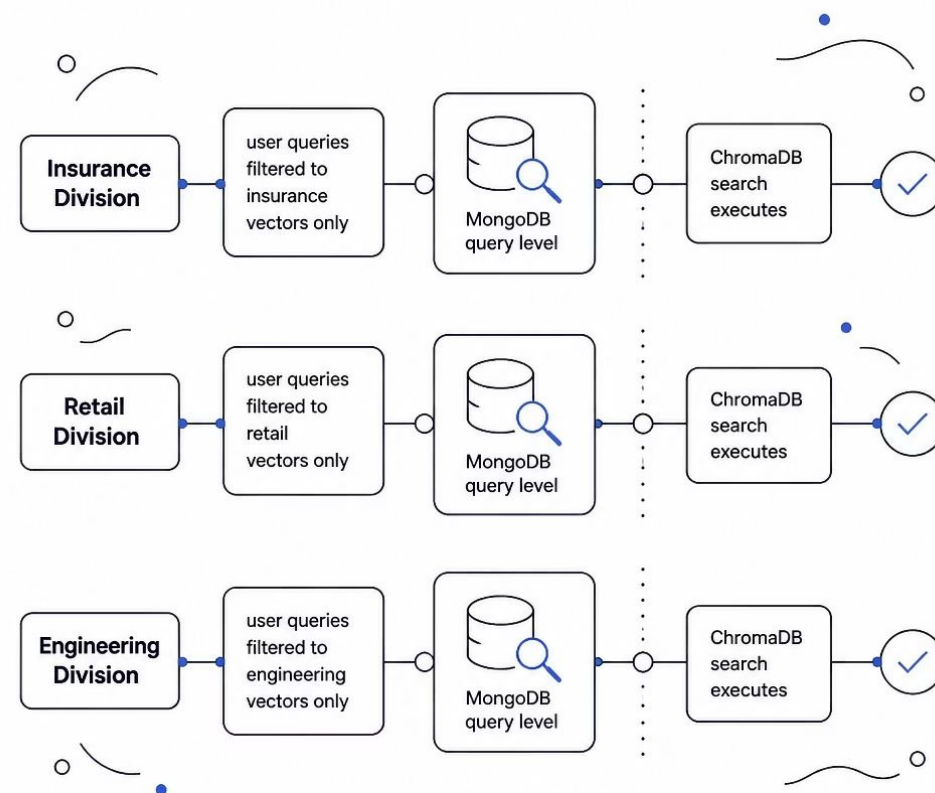
Security & User-Scoped Caching

Query-Level MongoDB Isolation

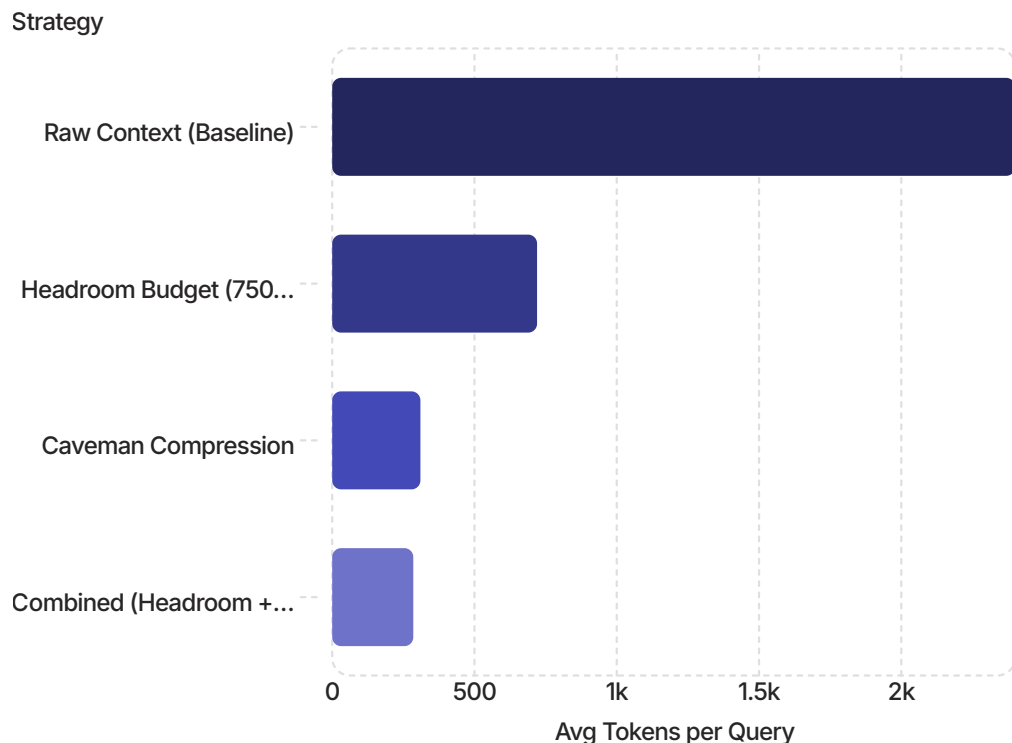
Every retrieval query is pre-filtered against the user's **division** and **businessLine** record in MongoDB. Insurance users never touch Retail vectors. Engineering never sees Finance metadata.

Semantic Cache Key: **user_email:query**

Identical queries from the same user hit the cache layer — bypassing embedding and LLM entirely. **90% latency reduction** on repeat queries.



Prompt Optimization Matrix



Two-Layer Token Discipline

Headroom Context Budget enforces a hard 750-token ceiling on retrieved context — truncating low-relevance chunks before they reach the LLM.

Caveman Output Compression strips verbose LLM preamble, markdown overhead, and redundant explanations — delivering only the answer payload.

Combined strategy achieves **88% token reduction** vs. baseline — directly translating to LLM cost savings at enterprise query volume.

Observability & RAGAS Scoring



Faithfulness

Measures whether generated answers are grounded in retrieved context — detecting hallucination at inference time.



Context Precision

Scores whether the most relevant chunks appear at the top of the retrieval window — optimizing token budget usage.



Answer Relevancy

Evaluates whether the final response directly addresses the user's query — closing the quality feedback loop.



Langfuse Trace Logging

Full trace-level observability: every span, token count, latency milestone, and RAGAS score logged in real time.



Business Impact

90%

LLM Cost Reduction

Via semantic caching, lazy embedding, and prompt compression at scale.

0

Hallucinations

RAGAS Faithfulness gating blocks ungrounded answers before delivery.

100%

Division Isolation

Query-level RBAC ensures zero cross-division data leakage by architecture.

<0.5s

Media Ingestion

Audio extraction from video via MoviePy — enabling real-time onboarding content pipelines.

✔ **Enterprise-ready from day one.** Multi-modal accessibility, absolute data isolation, and auditable AI — built for the compliance and scale demands of global enterprise onboarding.

